0.1 Summary

This thesis studied post-training weight quantization for large language models, with particular attention to the query, key, and value projections within grouped-query attention. Low-bit weight-only quantization can substantially reduce model storage and the memory traffic associated with inference. However, replacing full-precision weights with values on a coarse integer grid also perturbs the projection responses that participate in attention. The resulting distortion is not determined solely by the reconstruction error of each individual weight: it also depends on the activation multiplying that weight and on the interaction between the query and key projections.

To address these coupled effects, the thesis formulated two distinct refinement stages, Flip and ReFlip. Flip is an aggregate residual correction applied after round-to-nearest quantization. It revisits admissible integer levels and accepts a one-level change only when that change reduces the activation-weighted projection residual. Consequently, Flip does not require every modified weight to become individually closer to its full-precision value. A local reconstruction error may increase while the weighted errors within a projection row cancel more effectively, producing a smaller aggregate response error.

ReFlip addresses a different objective. It holds the quantized key projection fixed and updates only the quantized query projection to reduce a scalar query–key score surrogate. This query-only restriction makes the correction direction explicit and separates the score-level objective from the projection-level objective used by Flip. Parameter-level score sensitivities are aggregated to identify query dimensions with moderate correction potential, while Kneedle is used only to delimit a candidate region. Every proposed integer-level change must still be assessed against its exact change in the scalar surrogate loss. Thus, neither the candidate selection rule nor the greedy sequence of accepted changes is presented as a guarantee of global optimality.

The experimental evaluation considered both a narrowly controlled standalone setting and broader model-level metrics. The standalone analysis isolates the local quantity targeted by ReFlip and makes the behavior of the two refinement stages observable. The broader evaluation describes how quantized model variants behave on language-modeling and downstream tasks. Together, these two perspectives provide complementary evidence: the first examines the intended local correction mechanism, while the second indicates whether the resulting quantized variants remain competitive at model level. The evidence is interpreted descriptively and

within the boundaries of the evaluated settings.

0.2 Main Findings

The standalone case study provides the most direct evidence for the local mechanisms examined in this thesis. Relative to the initial quantized projection, the results show smaller activation-weighted projection residuals after Flip and smaller scalar query–key score mismatches after the subsequent ReFlip stage. This observed ordering is consistent with the objectives assigned to the two stages: Flip operates on an aggregate projection residual, whereas ReFlip operates on the residual of a scalar bilinear surrogate. The result does not establish that either stage minimizes a general attention loss, but it shows that the selected discrete corrections can be aligned with their respective local objectives.

The case study also illustrates why Flip and ReFlip should not be treated as interchangeable names for the same operation. Flip evaluates the effect of a candidate integer change through the projection residual induced by a representative activation vector. ReFlip begins from the Flip-refined state, fixes the key projection, and evaluates query-weight candidates through their exact effect on the current scalar score residual. A correction that is attractive under one objective is not necessarily attractive under the other. In particular, a ReFlip update may reduce the scalar surrogate loss while increasing the query projection residual. The observed improvements must therefore be understood as objective-specific rather than as uniform improvements of every reconstruction criterion.

The accepted corrections are sparse relative to a complete re-quantization of the selected projections. This behavior is compatible with the intended role of both methods as post-quantization refinements: they preserve the established quantization grid and alter only admissible integer levels whose evaluated loss changes are favorable. Sparsity alone does not establish efficiency on a particular hardware platform, but it indicates that the correction mechanism does not require replacing the quantization representation or optimizing every weight jointly.

At model level, the evaluated quantized variants display task-dependent differences in perplexity and downstream accuracy. Refined variants are competitive in several comparisons, while the relative ordering varies across models, baselines, and metrics. These results are best interpreted as descriptive evidence that the local corrections can coexist with multiple quantization baselines. They do not, by themselves, isolate the contribution of every design choice or demonstrate that a reduced scalar score mismatch necessarily causes an improvement on a downstream task. Such a causal conclusion would require broader controlled ablations and repeated evaluations.

Taken together, the findings support a careful conclusion. Aggregate activation-weighted residuals and scalar query–key score residuals provide meaningful alternatives to individual weight reconstruction error when selecting discrete post-quantization corrections. Flip and ReFlip offer two complementary ways to exploit these objectives. Their usefulness, however, remains conditional on the chosen representative statistics, candidate-selection policy, quantization configuration, and evaluation setting.

0.3 Limitations

The first limitation concerns the representative activation vector. Both the activation-weighted Flip objective and the scalar ReFlip surrogate summarize a broader activation distribution through a single representative vector. This makes the discrete loss changes tractable and interpretable, but it cannot preserve every token-specific or sequence-specific projection response. A correction that is favorable for the representative vector may have a smaller benefit, no benefit, or an adverse effect for other activations. The conclusions therefore apply to the evaluated representative objective rather than to all possible inputs.

Second, the scalar query–key score surrogate is not the full attention matrix. It considers an unscaled bilinear interaction for a selected query head and its associated key head under grouped-query attention. It does not model all token-to-token logits, softmax normalization, causal masking, interactions among sequence positions, or the subsequent multiplication by value representations. Reducing this scalar mismatch is a focused local objective; it is not equivalent to preserving the complete attention operation.

Third, the optimization procedures are local and greedy. Kneedle identifies a region of moderate dimension-level sensitivity, but a knee in the sorted sensitivity curve does not certify that the selected region contains the globally best corrections. Within that region, integer-level candidates are accepted sequentially according to their current exact local loss change. Earlier accepted changes alter the residual used to evaluate later candidates, and the resulting path may differ from a joint optimum. Neither Kneedle selection nor greedy acceptance provides a global optimality guarantee.

Fourth, the standalone analysis has a deliberately narrow scope. It examines one selected attention layer, one grouped-query attention group, a small collection of query heads, and a single representative activation statistic. This scope is valuable for inspecting the correction mechanism without conflating it with the many transformations in a complete model, but it limits generalization across layers, head groups, architectures, calibration distributions, and quantization settings. Broader repeated studies are required before drawing claims about consistent behavior throughout an entire model.

Finally, the selected model-level metrics cover only part of the deployment problem. Perplexity and downstream accuracy do not directly characterize long-context behavior, generation quality, robustness, latency, memory bandwidth, energy consumption, or hardware-specific inference throughput. Likewise, the present comparisons do not constitute a complete ablation of the representative statistic, sensitivity aggregation, candidate budget, or knee-selection policy. These dimensions remain necessary for a more comprehensive assessment.

0.4 Future Work

Future evaluation should extend both the controlled and model-level settings. Repeating the standalone analysis across layers, GQA groups, query heads, calibration samples, and model families would reveal whether the observed correction patterns are stable or localized. A larger task suite, long-context benchmarks, repeated trials, and deployment measurements would provide a stronger account of quality, variance, memory behavior, and inference efficiency. Controlled ablations should separately examine the representative-vector estimator, the sensitivity aggregation rule, the Kneedle configuration, and the correction budget.

The correction strategy can also be generalized. ReFlip currently uses exact scalar-surrogate loss changes inside a heuristically selected candidate region and modifies only the query projection while holding the key projection fixed. Future work may study alternating query and key corrections, group-level objectives that aggregate several query heads sharing one key head, and objectives defined over multiple representative activations. Joint or beam-search procedures could also be compared with greedy acceptance to measure the quality lost through local decisions. These extensions must retain explicit integer constraints so that an apparent continuous improvement corresponds to a valid change on the quantization grid.

A particularly important direction is to formulate a value-projection correction through a fixed-attention value-output objective rather than through a query–key score. This objective evaluates the output after value mixing while deliberately holding the attention matrix fixed; it therefore targets the value path without implying end-to-end optimization of the complete attention operation. Let the full-precision target output and a quantized value-path output be

$$O^\star = A^\star V^\star, \qquad \widehat{O} = \overline{A}\,\widehat{V}, \qquad L_V = \frac{1}{2}\left\|\widehat{O} - O^\star\right\|_F^2,$$

where $A^\star$ is the full-precision attention matrix, $\overline{A}$ is the attention matrix held fixed while evaluating a value-only correction, $V^\star = XW_V^\top$, and $\widehat{V} = X\widehat{W}_V^\top$. The fixed matrix $\overline{A}$ may be chosen according to the evaluation objective, but that choice must be stated explicitly because it determines which attention-path error is attributed to the value projection.

For a proposed one-level integer change $\Delta z_{ab} \in \{-1, +1\}$ in a quantized value weight, the corresponding dequantized weight change is

$$\Delta\widehat{W}_{V,ab} = \alpha_{a,g(b)}\Delta z_{ab}.$$

The following integer-level loss-change derivation is exact under the local correction setting in which $\overline{A}$, the quantization scale $\alpha_{a,g(b)}$, the associated zero-point, and every weight other than the selected value weight are held fixed throughout evaluation of the candidate. This produces a structured value-response change $\Delta\widehat{V}$ and an output change

$$\Delta\widehat{V} = X\left(\Delta\widehat{W}_V\right)^\top = \alpha_{a,g(b)}\Delta z_{ab}\,(Xe_b)e_a^\top, \qquad \Delta\widehat{O} = \overline{A}\,\Delta\widehat{V}.$$

Here, e_a and e_b denote standard basis vectors for the output and input dimensions, respectively. The rank-one form makes the effect of a legal integer move explicit before it is propagated through the fixed attention matrix. Writing the current output residual as $R_O = \widehat{O} - O^\star$, the exact loss change for that admissible integer move is

$$\Delta L_V = \left\langle R_O, \Delta\widehat{O}\right\rangle_F + \frac{1}{2}\left\|\Delta\widehat{O}\right\|_F^2.$$

An output-aware value correction could accept the candidate only when $\Delta L_V < 0$, update R_O, and then evaluate the next candidate against the new residual. This formulation preserves the discrete, exact-loss-change principle used by Flip and ReFlip while aligning the value-weight objective with the representation produced after attention mixing. As with the query-only formulation, a greedy sequence of value corrections would remain a local procedure rather than a guarantee of global optimality.

More broadly, future work should investigate how projection-level, score-level, and output-level objectives can be combined without obscuring their distinct meanings. A unified method could allocate correction budgets according to the residual that each projection can directly influence, while retaining transparent acceptance criteria for every integer-level adjustment. Such a formulation would provide a principled path from local quantization-error correction toward preservation of the complete attention computation.